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CROSSDEM: FROM VOICE TO VECTOR **AI-DRIVEN ANALYSIS OF THE VARIETIES OF DEMOCRACY**

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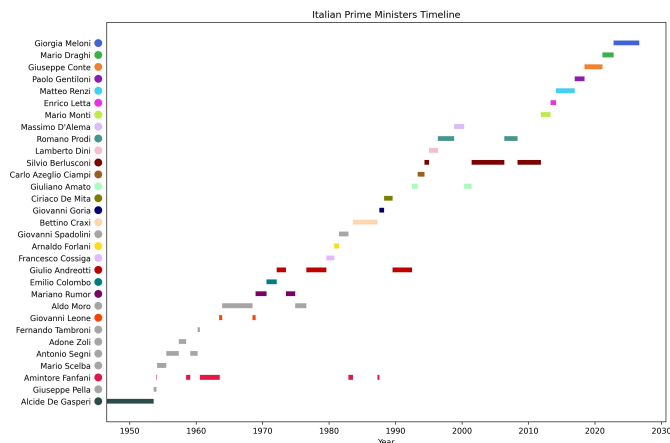
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<https://github.com/mhetacc/crossdem>



LOOK AGAIN AT THAT DOT. THAT’S HERE. THAT’S HOME. THAT’S US.

ON IT EVERYONE YOU LOVE, EVERYONE YOU KNOW, EVERYONE YOU EVER HEARD OF, EVERY HUMAN BEING WHO EVER WAS, LIVED OUT THEIR LIVES.

THE AGGREGATE OF OUR JOY AND SUFFERING, THOUSANDS OF CONFIDENT RELIGIONS, IDEOLOGIES AND ECONOMIC DOCTRINES, EVERY HUNTER AND FORAGER, EVERY HERO AND COWARD, EVERY CREATOR AND DESTROYER OF CIVILIZATION, EVERY KING AND PEASANT, EVERY YOUNG COUPLE IN LOVE, EVERY MOTHER AND FATHER, HOPEFUL CHILD, INVENTOR AND EXPLORER, EVERY TEACHER OF MORALS, EVERY CORRUPT POLITICIAN, EVERY “SUPERSTAR”, EVERY “SUPREME LEADER”, EVERY SAINT AND SINNER IN THE HISTORY OF OUR SPECIES LIVED THERE—ON A MOTE OF DUST SUSPENDED IN A SUNBEAM.

— CARL SAGAN, *PALE BLUE DOT*, 1994

Abstract

In the past decades, since the introduction of the Internet of Things (IoT) paradigm, the number of connected devices has grown exponentially. These applications now dominate our everyday life across multiple domains, such as smart homes, healthcare or industrial automation.

This growth has proceeded hand in hand with the development of various protocols to support different requirements. However, this technological variety introduces significant architectural fragmentation, which may limit interoperability. The nature of some IoT applications has raised significant security and privacy concerns, especially with regard to sensitive data or critical infrastructures, where a lack of security measures and cyber-attacks could have severe consequences. Therefore, a risk analysis is essential to identify vulnerabilities and design defence effective strategies.

In this thesis, we present a multi-protocol IoT framework designed to address the challenge of protocol fragmentation by providing a unified, protected communication environment. The work focuses on the design, implementation and security analysis of the two most widely adopted IoT protocols: MQTT and CoAP. This framework is applied to the specific use case of smart agriculture, with a focus on monitored and automated greenhouses. The core of the study involves addressing potential cyber threats by implementing appropriate security mechanisms, such as authentication, authorization and encryption, to mitigate risks without compromising the efficiency required by agricultural sensors.

The work demonstrates the challenges of integrating heterogeneous IoT protocols within a secure framework, highlighting the importance of addressing both interoperability and security.

This project was carried out during the internship at M3I S.r.l., an IoT solutions company based in Padova.

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¹A speech is considered "right-leaning" if it pertains a right-leaning Prime Minister.

²When speeches for each leaning are present. Otherwise only two, one, or no set at all can be extracted from that 5-year time frame.

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³Top-1000 words means pooling the one thousand most used words of a specific Prime Minister. Stopwords are removed and the lexicon is lemmatized.

⁴Each speech counts as a document.

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List of Acronyms

MQTT	Message Queuing Telemetry Transport
CoAP	Constrained Application Protocol
IoT	Internet of Things
BLE	Bluetooth Low Energy
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
NB-IoT	Narrowband Internet of Things
AES-CCM	Advanced Encryption Standard - Counter with CBC-MAC
MAC	Message Authentication Code
TCP	Transmission Control Protocol
mTLS	Mutual Transport Layer Security
M2M	Machine to Machine
IAM	Identity Access Management
HA	High Availability
IDS	Intrusion Detection Systems
ReBAC	Relationship-Based Access Control

1

Introduction

Politicians are some of the most influential people in history: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to shape society. Shapiro and Page (1984) [1] showed how nearly all of the 20th century U.S. Presidents sought to steer public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, **speeches** play a vital role in the functioning of society [2]: leaders depend on the verbal power to persuade people [3].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. (2021) [4] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump⁵ attacking the legitimacy of the 2020 US presidential election. While they found no evidence indicating that elites can erode democratic norms easily, elite rhetoric can shape people's views about core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Speeches can be differentiated between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. Social networks are especially interesting since they afford politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57,000 times between May 2009 and January 2021 [5], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [6].

While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [7,8]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [9,10], and there have been attempts to predict election results with analyses on data from

⁵Former president when Clayton et al. wrote their paper, current president at the time of writing this Master Thesis.

social media. In Belcastro et al.'s 2022 study [11] a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Opposite results can be seen in Rita et al.'s work (2023) [12], which measures sentiment polarity on Twitter, and concludes that tweets sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions.

Silva et al. (2022) [13], on the other hand, managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

Circling back to the matter of speeches, external speeches often need to be scraped. Social networks can simplify this process through their exposed APIs and the availability of community-developed tools such as Python's library `ytp-dl`⁶. These resources open the possibility to automate speech collection (section 3.3.2).

On the other hand, internal speeches are usually already available in easy to access, *processing-ready* format⁷, yet their technical and codified nature makes their interpretation harder for both humans and automated agents. Moreover, until 1985 the Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [14, 15].

Broadly speaking, this project aims to compile a corpus of political speeches and subject it to linguistic analyses, progressing from classical NLP metrics such as word frequency and textual complexity to LLM-assisted classification tasks (hate-speech detection and target identification amongst them). This is why I ultimately chose to gather a corpus of **external speeches** rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung and Kim's controlled study (2025) [16] scaled the vocabulary of a set of language models from 24,000 to 196,000 tokens while holding data, computation, and optimization unchanged. They discovered that the models are disproportionately optimized on the part of language that occurs most frequently, meaning they are far better at understanding words and patterns that appear more often. This suggests that internal speeches are less understandable to AI agents.

Moreover, Birkenmaier and Lechner (2025) [17] made a similar dataset choice: they measured politicians' public personality traits using computational text analysis, and their corpus comprises data from interviews, social media posts, and parliamentary speeches. Specifically: "interview statements provide one of the most natural and authentic sources of politicians' interactions and self-presentation to the public", making them one of the best data sources for political speeches [18, 19].

Let us now shift the **focus to the analyses** themselves.

There's a widespread view that populism is on the rise, from the United States and Turkey to India and Hungary [20–23], a development that has generated considerable concern because, in its most radical

⁶Privacy-focused media downloader API for Linux VPS deployments: <https://pypi.org/project/ytp-dl/>

⁷For example, the Italian Senate's website contains a database with every (recent) seating in HTML format. An example of seating n.1, March 23 2018, can be seen at the following link: <https://www.senato.it/show-doc?tipodoc=Sindisp&leg=18&id=1066811>

and authoritarian form, populism threatens democracy [24], polarizes societies [25], and erodes trust in experts [26].

Moreover, the progressive simplification of political discourse [27] has attracted increasing attention, as it allegedly favours populist politicians, particularly those on the far right. Many studies tried to explore these claims, with mixed results: Decadri and Boussalis (2019) [28] used text analytic techniques to inspect parliamentary speeches given by the members of nine Italian parties. Their results suggest that populist ideology, electoral strategy, and party membership influence legislators' language complexity, and that language simplicity might be thought of as a feature of populist communication.

On the other hand, McDonnell and Ondelli in their 2020 study [29] investigate the linguistic simplicity of four right-wing populists compared to their principal opponents in the United States, France, United Kingdom, and Italy. Contrary to expectations, they find that Donald Trump was only slightly simpler than Hillary Clinton, while Nigel Farage in the UK and Marine Le Pen in France were more complex than their main rivals, and Italy's Matteo Salvini was simpler on some measures but not others, concluding that the relationship between populism and simplicity should not be taken for granted.

Similar results were found by Rebecca C. Kittel in 2025 [27] after analyzing German parliamentary debates from January 1991 to September 2021. Their findings show that right-wing populist actors use the most complex language, while left-wing populists' language complexity is average. At the same time, the study finds that overall complexity decreased significantly in the German parliament over time, confirming at least partially the claims about "progressive simplification".

Other studies shifted the focus from textual complexity to the emotiveness⁸ of speech. Wang et al. (2026) [31] explore the sound of populism by integrating classic Linguistic Inquiry and Word Count with a fine-tuned RoBERTa model. Their findings reveal that populist rhetoric consistently features a direct, assertive "sound" that forges a connection with "the people" and constructs a charismatic leadership persona. Notably, right-wing populism and people-centrism exhibit a more emotionally charged discourse, resonating with themes of identity, grievance, and crisis, in contrast to the relatively restrained emotional tones of left-wing and anti-elitist expressions.

Bliuc et al.'s work (2026) [32] highlights how anger consistently emerges as the most powerful emotional correlate of populist attitudes, followed by fear, resentment, and nostalgia, which agrees with Laura Alonso-Muñoz and Casero-Ripollés's 2023 study [33]. Their findings show that fear, uncertainty and resentment are the emotions most frequently used by European populist parties and leaders⁹.

Caiani and Di Cocco (2023) [34], on the other hand, try to draw the connection between emotion and populism using machine learning: focusing on Italy as a case study, they systematically investigate the intensity and trends of specific emotions in political discourses of all Italian political parties over the last 20 years. Their findings confirm that populist politicians tend to leverage more emotional appeals than non-populist parties. However, there is an increase in the use of these appeals overall, especially in terms of negative emotions. Most notably, different types of emotions are mobilized by different types of populisms: right-wing populists mainly use negative emotions, while left-wing populists employ positive emotional appeals.

⁸ *Emotive*, adjective: 1. of or relating to the emotions; 2. appealing to or expressing emotion; 3. causing strong emotions often in support of or against something [30].

⁹ Alonso-Muñoz and Casero-Ripollés for their 2023 study built a corpus of tweets from four European political parties: Podemos, 5 Star Movement, National Front, and the UKIP [33].

Turning now to the effects populism has on society, polarization may be the one most consistently observed [35,36]. A lot of research has been done on the risks posed by a polarized society, such as Benson’s (2023) [37], which argues that polarization’s epistemic harms are best located in its tendency to reduce the diversity of perspectives utilized in a democratic system and in how this weakens the system’s ability to identify and address problems of public concern.

Rostbøll (2024) [38] argues that “it is widely agreed that the increased polarization many countries experience is bad for democracy”, and recommends a systemic approach to assess the democratic implication of polarization, which analyzes both the effects of polarization at different sites and on democracy as a composite whole.

Schedler (2023) [39] frames political polarization as a form of public conflict that aims to destroy the basic democratic trust. Citizens living in a polarized society fear for the subversion of democracy instead of their economic well-being, physical safety, or ways of life. They then argue that the institutional implications of such fears are dramatic, spelling the end of democratic consolidation.

Other literature argues that affective polarization contributes to democratic erosion by increasing partisan loyalty and decreasing the importance citizens give to democratic procedures. More specifically, the strength of partisanship has also been found to be associated with “partisan double standard” [40] or “democratic hypocrisy” [41], that is, the willingness to overlook democratic violations by one’s own party.

This Master Thesis will explore the three verticals just discussed (textual complexity, polarization, and emotiveness) from a computational point of view.

The dataset consists of over five thousand annotated speeches delivered by Italian Prime Ministers between 1945 and 2025. Section 3.3 outlines how the corpus was built, and section 3.3.3 describes in detail how it was annotated to address the topic of “emotiveness”: every speech is classified by its degree of aggressiveness, hate speech or negativity, as well as whether it targets a specific group or not. Possible target groups are: political adversaries, gender minorities, ethnic minorities, and religious minorities.

Section 4.1 showcases all the analyses performed on the matter of textual complexity, section 4.2 does so on the matter of polarization, while section 4.3 explores the matter of emotiveness.

Lastly, in section 5 the results are discussed as well as possible future work.

1.1 RESEARCH QUESTIONS

Before delving into the research questions, it is important to specify the term *hostility*.

In the pursuit of measuring emotiveness, I measured three qualities of speech (hate speech, aggressiveness, and negativity), and I classified whether a speech is targeting a group or not (political adversaries, gender minorities, ethnic minorities, and religious minorities).

I define the term *hostility* as the compound metric that incorporates all three aforementioned qualities¹⁰.

¹⁰To clarify: if a speech has high negativity and high aggressiveness, it is a speech with medium to high hostility.

1. Textual Complexity:

- 1.1. Does it have a correlation with the quality of democracy?
- 1.2. Does it evolve over time? How?
- 1.3. Does it change amongst different political leanings?
- 1.4. Does lexical complexity evolution exhibit some common patterns amongst different Prime Ministers?

2. Polarization:

- 2.1. Can we identify symptoms of polarization?
 - 2.1.1. Per Prime Minister;
 - 2.1.2. Per political leaning;
 - 2.1.3. Per year period;
- 2.2. Do the symptoms evolve over the years? Do the evolution patterns change depending on the political leaning?
- 2.3. Do the symptoms have a correlation with the quality of democracy?

3. Emotiveness:

- 3.1. Can we identify hate speech?
- 3.2. Can we identify negativity?
- 3.3. Can we identify aggressiveness?
- 3.4. Do emotionally charged speeches evolve over the years?
- 3.5. What emotions are used the most? By which leaning? In which periods?
- 3.6. Does hostility have a correlation with the quality of democracy?
- 3.7. What Prime Minister is more hostile?
- 3.8. What political leaning is more hostile?
- 3.9. What period is more hostile? Is hostility evolving? How?
- 3.10. Target:
 - 3.10.1. Which group is more often targeted?
 - 3.10.2. Which political leaning, if any, targets groups more often?
 - 3.10.3. Which Prime Minister, if any, targets groups more often?
 - 3.10.4. Which time periods, if any, have more speeches that target a group?

2

State of the Art

2.1 TEXTUAL COMPLEXITY

Text complexity or readability refers to how difficult a text is to understand [42], influenced by linguistic factors such as word choice (e.g., “utilize” vs. “use”), sentence structure (complex vs. simple), and content type (academic vs. children’s books) [43,44].

Common comprehension metrics are Flesch Reading Ease (FRE) [45], which considers the average sentence length and syllables per word, and outputs a value between zero (hard to read) and one hundred (easy to read); Dale-Chall [43], which measures a raw score taking in consideration a list of 3000 words that groups of fourth-grade American students could reliably understand, considering any word not on that list to be difficult; and SMOG [46], which uses intuitive features like sentence length and word complexity.

Another way to measure the complexity is by measuring the **lexical diversity**, which is the ratio of different unique word types compared to the total number of words. In a sense, it measures the vocabulary richness of a sample.

The most common metric is the Type-Token Ratio, obtained by simply dividing word types over tokens. A TTR of 0.5 means that half of the words in the sample appear only once.

A more refined metric was introduced by McCarthy in 2005: it is the measure of textual lexical diversity (MTLD) [47]. It uses a sequential algorithm to count tokens until lexical repetition forces the ratio below a fixed threshold, reducing text-length bias, and it’s widely applied in computational linguistics for corpus profiling, tracking writing development, and evaluating machine translation quality. MTLD has become a standard tool in applied linguistics and computational text analysis, and it is often used as a measure of lexical diversity, lexical complexity, or lexical sophistication [48].

Let’s now have a look at the recent research.

Bestgen (2024) [49] published a methodological review on the subject of lexical diversity, (type–token ratio), MTTRRS (mean type–token ratio in random samples), MATTR (moving-average type–token ratio), MSTTR (mean segmental type–token ratio), MTTRSS (mean type–token ratio in sequential samples), and MTLD (measure of textual lexical diversity). Its findings show that for a local estimation, MATTR and MTLD are recommended. While MTLD appeared to be slightly more sensitive to length than MATTR, the differences were insufficient to declare one superior to the other.

Velasco and Roque (2025) [50] explored the role of text complexity in language model pretraining. In their study, they measured how complexity affects language modeling across model sizes, whether useful representations can be learned from simpler text alone, and how text complexity in pretraining influences

downstream language understanding. They used a subset of *FineWeb-Edu* [51], then simplified by prompting *Llama 3.1 8B* [52] at the paragraph level.

To get a rough idea of what the simplified texts look like, see the following example:

Original: *Your comment really helped me feel better the most. I was sitting in my office, feeling so bad that I didn't say how inappropriate and out of line his comments were, and this helped.*

Simplified: *Your comment really helped me feel better. I was feeling bad because I didn't speak up when someone made inappropriate comments.*

Since LLM-based scoring is computationally costly, they used Flesch Reading Ease to estimate readability, and Type-Token Ratio to measure lexical diversity.

Amadori et al. (2025) [53] examined the linguistic complexity of content produced by influential users on Twitter across three globally significant and contested topics: COVID-19, COP26, and the Russia–Ukraine war. Their findings show significant differences across account type, political leaning, content reliability, and sentiment. Additionally, profiles producing more negative and offensive content tend to use more complex language.

They used three metrics: Flesch readability score; Yule's K measure of lexical richness [54], which quantifies how repetitive or diverse the vocabulary is, and it is largely independent of the overall length of a text; and gzip-based compression complexity, which assesses text repetitiveness by measuring its compression ratio, an approach already explored in previous studies [55–57].

Lastly, Fredrick and Craven (2025) [58] compared AI-generated texts (via ChatGPT) and student-written essays in terms of lexical diversity, syntactic complexity, and readability, using metrics such as Type-Token Ratio and Flesch-Kincaid. Results indicate that while ChatGPT produces texts with greater lexical diversity and syntactic complexity, the outputs are overall less readable.

They chose a range of linguistic complexity metrics: for lexical diversity they used TTR and MTLD, for syntactic complexity MLT and DC/T, for readability Flesch-Kincaid and Gunning-Fog, and lastly they also measured lexical sophistication.

The main metrics are summarized in table 2.1.

For the purposes of this project, I decided to limit the scope to one lexical complexity metric. MTLD was chosen due to its wide adoption in literature and its robustness to text length: collected speeches (section 3) vary in length between one another, going from a couple of lines to more than twenty thousand words.

Metric	What it measures	Basis	Output
TTR	Lexical diversity	Ratio of unique word types to total tokens	Raw ratio (0–1), length-sensitive
MTLD	Lexical diversity	Mean length of word sequences maintaining a TTR above a fixed threshold (0.720)	Words per factor, robust to text length
Flesch Reading Ease	Sentence/word-length readability	Average sentence length + average syllables per word	0–100, higher = easier
Dale–Chall	Vocabulary familiarity	% of words outside a ~3,000-word familiar-word list + average sentence length	US grade level
SMOG	Polysyllabic density	Count of 3+ syllable words across a fixed sentence sample	US grade level
Yule’s K	Lexical diversity	Statistic on word-frequency distribution, weights repeated word types more heavily	Constant K, lower = more diverse
Gzip Compression	Information-theoretic complexity	Compressed-to-original size ratio via gzip; more compressible text = more redundant/predictable	Compression ratio, lower = more predictable
Flesch-Kincaid Grade Level	Sentence/syllable-density readability	$0.39 \times \text{words/sentence} + 11.8 \times \text{syllables/word} - 15.59$	US grade level
Gunning Fog Index	Polysyllabic-word density	$0.4 \times (\text{words/sentence} + 100 \times \text{complex-words/words})$	US grade level
MLT	Syntactic complexity (elaboration)	Total words / total T-units	Words per T-unit
DC/T	Syntactic complexity (subordination)	Total dependent clauses / total T-units	Dependent clauses per T-unit

Table 2.1: Comparison of common textual complexity/readability metrics

2.2 POLARIZATION

- What is

Political polarization is the divergence of political attitudes away from the center, towards ideological extreme [59–61]. Scholars distinguish between ideological polarization (differences in the policy positions) and affective polarization (an emotional dislike and distrust of political out-groups) [62], and between elite polarization (polarization of the political elites, such as elected official), and mass polarization, that focuses on the polarization of the general public [63–65].

- Common metrics
- State of the art: what has been done in the field, what are the gaps, what are the limitations



2.3 SENTIMENT ANALYSIS

- What is
- Common metrics
- State of the art: what has been done in the field, what are the gaps, what are the limitations



3

The Corpus

3.1 SIMILAR CORPORA

3.2 MY CORPUS

- Structure
- Content
 - prime ministers
 - transcriptions
 - annotations

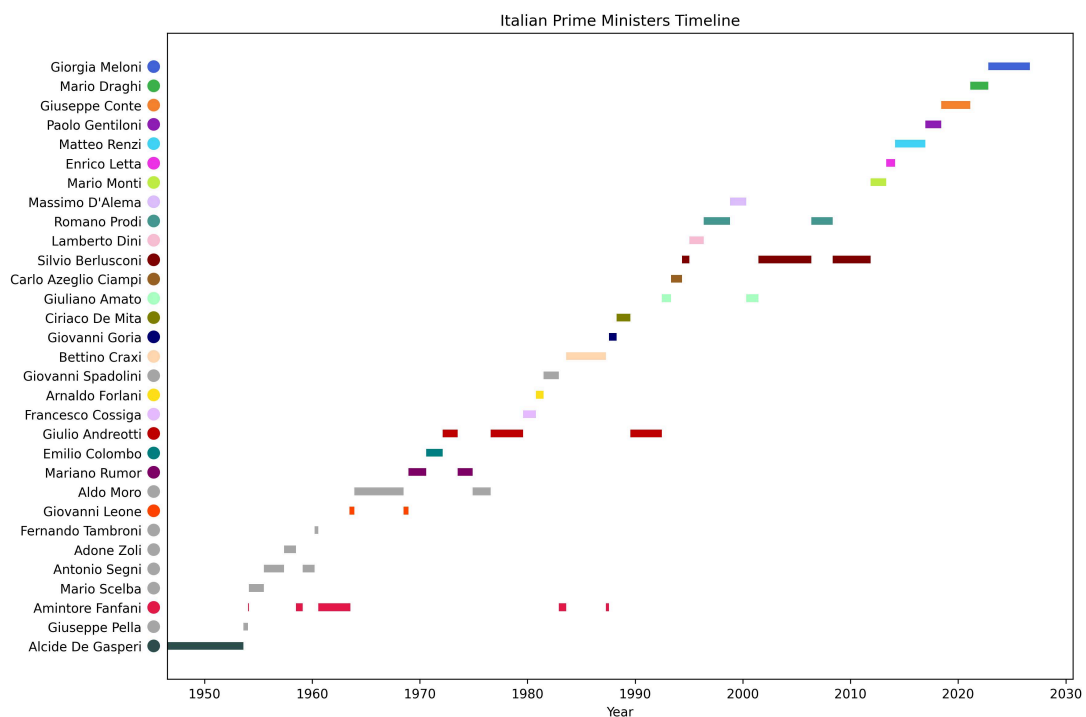
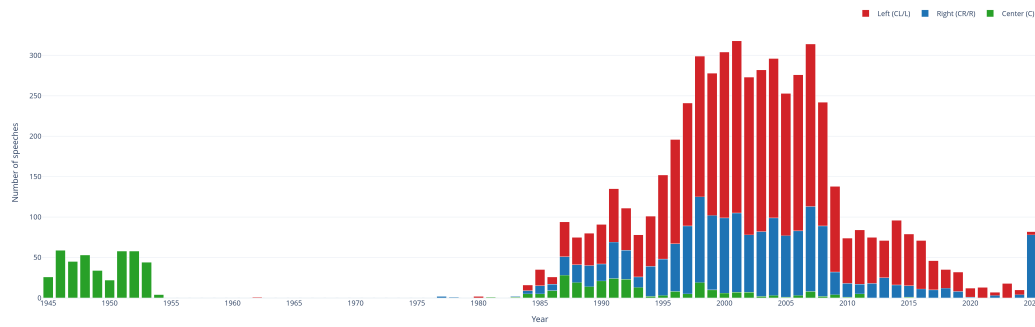
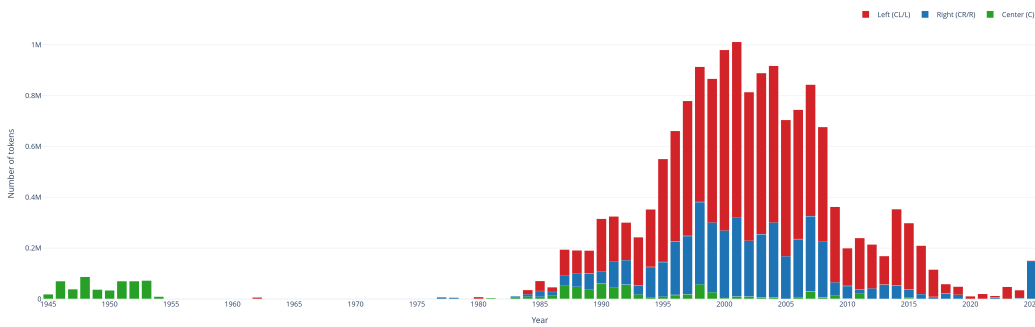


Figure 3.1: Italian Prime Ministers from 1946 to 2025. In grey Prime Ministers not included in the corpus.



(a) Dataset: words. The graph show strong data imbalances: years between 1995 and 2010 contain most of the data, and left-leaning Prime Ministers are overrepresented.



(b) Dataset: tokens. Tokenization done via `spacy.load("it_core_news_sm")`.

Figure 2: 2a show the number of words collected for the dataset over the span of eighty years. In green are words pertaining to **center-leaning** Prime Ministers, in red are words pertaining to **left-leaning** Prime Ministers, and in blue are words pertaining to **right-leaning** Prime Ministers. 2b, on the other hand, shows the same data but in terms of tokens instead of whole words.

3.3 BUILDING THE CORPUS

3.3.1 DE GASPERI

Thanks to Tonelli et. al. [66] for the De Gasperi corpus.

3.3.2 SCRAPING

- Scraping
 - Meloni YouTube
 - Radio Radicale
- Speech-to-text via OpenAI *Whisper*

3.3.2.1 SCRAPING SANITY CHECK

Is the corpus correct? Yes, check with n-gram centroid.

3.3.3 CORPUS ANNOTATION

- Choosing how to annotate the corpus
 - Choosing the right LLMs



- Pipeline

3.3.4 V-DEM



Results

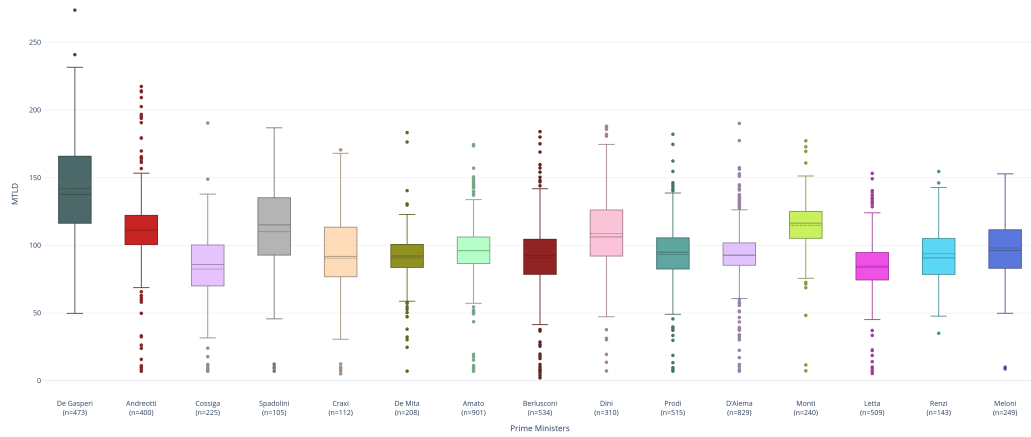
4.1 TEXTUAL COMPLEXITY

4.1.1 MEASURE OF TEXTUAL LEXICAL DIVERSITY

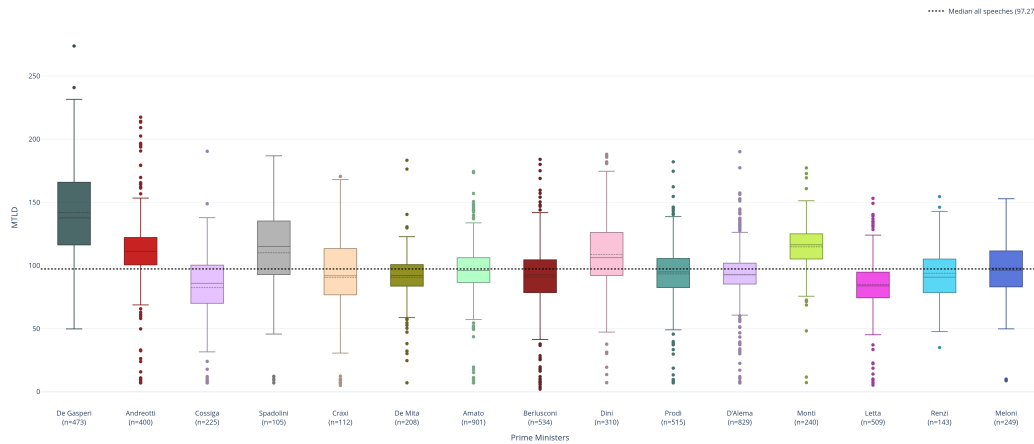
Measure of Textual Lexical Diversity (MTLD) is a metric used to assess the diversity of vocabulary in a text. It calculates the average length of sequences of words that maintain a certain level of lexical diversity, providing insights into the richness and variety of language used in the text. yada yada

4.1.2 PRIME MINISTER'S MTLDS

4.1.2.1 BOXPLOT OF MTLD VALUES FOR EACH ITALIAN PRIME MINISTER



(a) Boxplot of MTLD values for each Italian Prime Minister, without overall median.

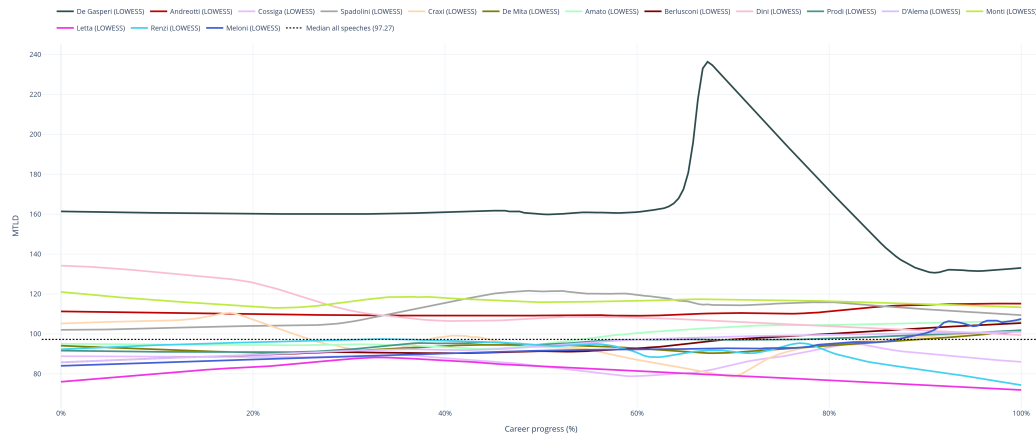


(b) Boxplot of MTLD values for each Italian Prime Minister, with overall median.

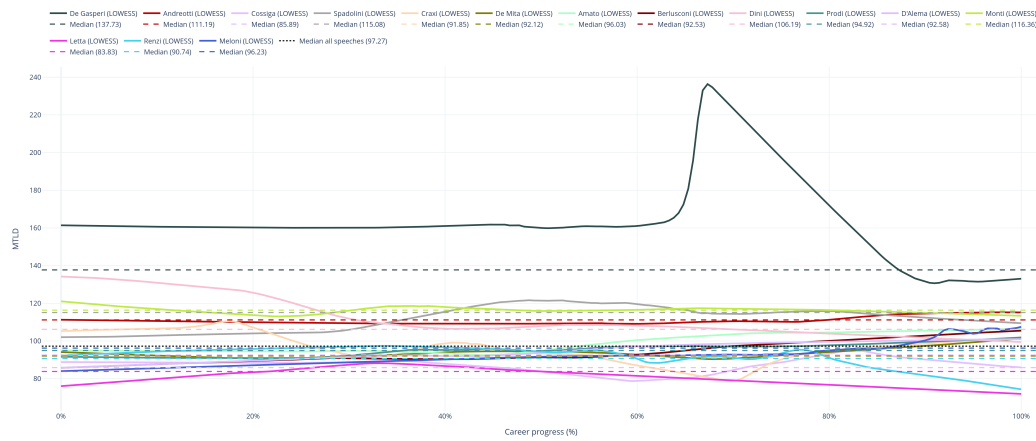
Figure 3: 3a and 3b show MTLD values for each Italian Prime Minister, ordered by their first time in office. This allows for not only a direct comparison between Prime Ministers, but also a comparison of the evolution of MTLD values over time. Prime Minister with fewer than 100 speeches are omitted.

4.1.2.2 EVOLUTION OVER TIME OF MTLD VALUES FOR EACH ITALIAN PRIME MINISTER

- what is LOWESS



(a) LOWESS curve of MTLT values for each Italian Prime Minister, without overall medians.

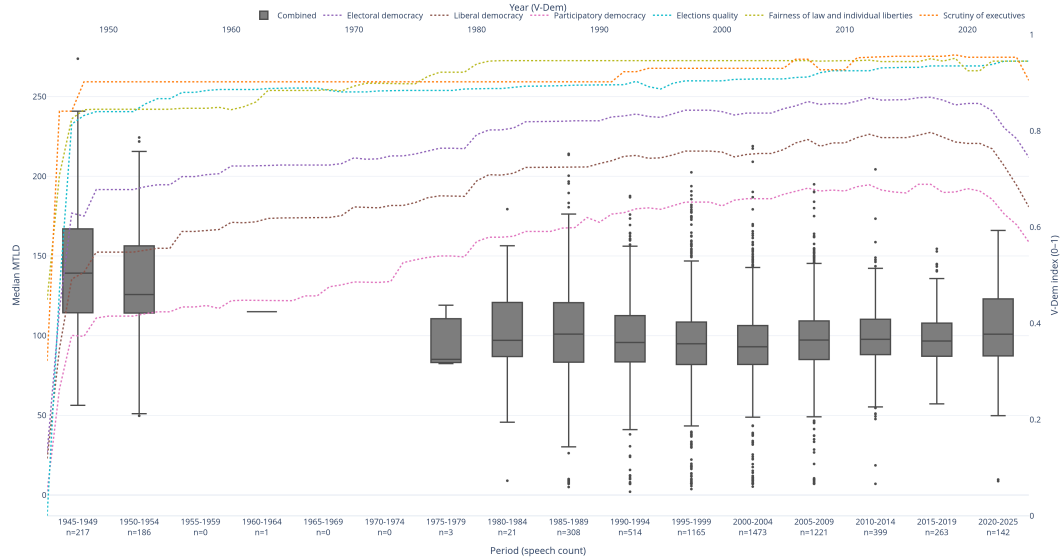


(b) LOWESS curve of MTLT values for each Italian Prime Minister, with overall medians.

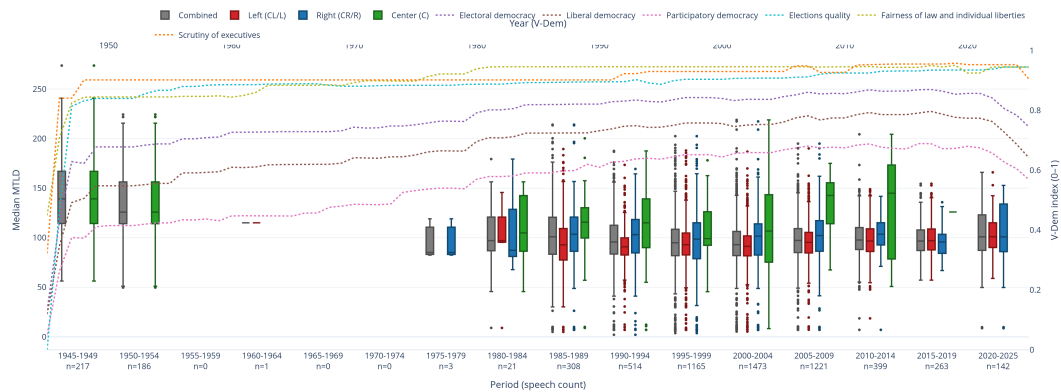
Figure 4: 4a and 4b show MTLT values for each Italian Prime Minister over their careers span. The LOWESS curve allows was chosen to aid in the interpretation of the values, and the overall medians allow for a comparison between Prime Ministers. Prime Minister with fewer than 100 speeches are omitted.

4.1.3 VDEM

4.1.3.1 5 YEARS



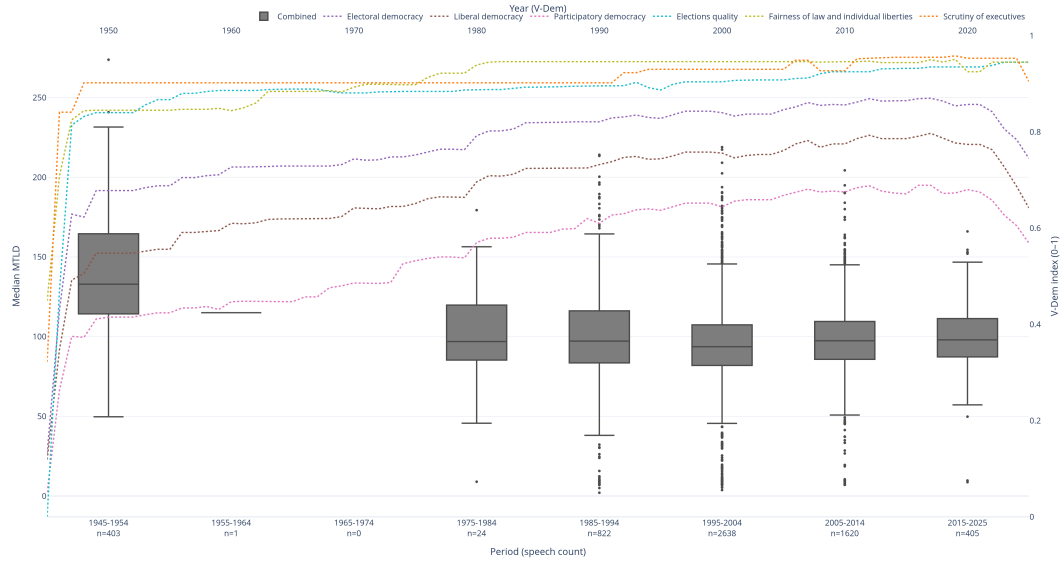
(a) MTLD values in a Boxplot by 5-year periods, with VDEM indices. Each grey box combines all speeches for all political leanings in that 5-year period.



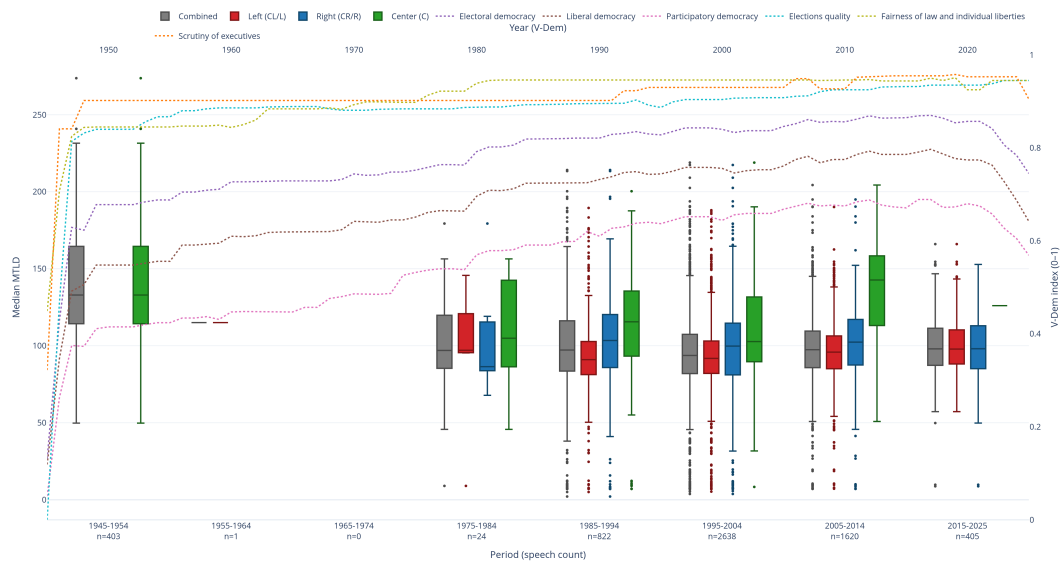
(b) MTLD values in a Boxplot by 5-year periods, with VDEM indices. As above, each grey box combines all speeches for all political leanings in that 5-year period, while the colored boxes represent respectively center-leaning (in green), left-leaning (in red), and right-leaning (in blue) MTLD values in that 5-year period.

Figure 5: MTLD values in a Boxplot by 5-year periods. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2019, which an abrupt inversion in the 2020-2025 period. If this is a correlation with the inflection of some VDEM indices, it is not clear.

4.1.3.2 10 YEARS



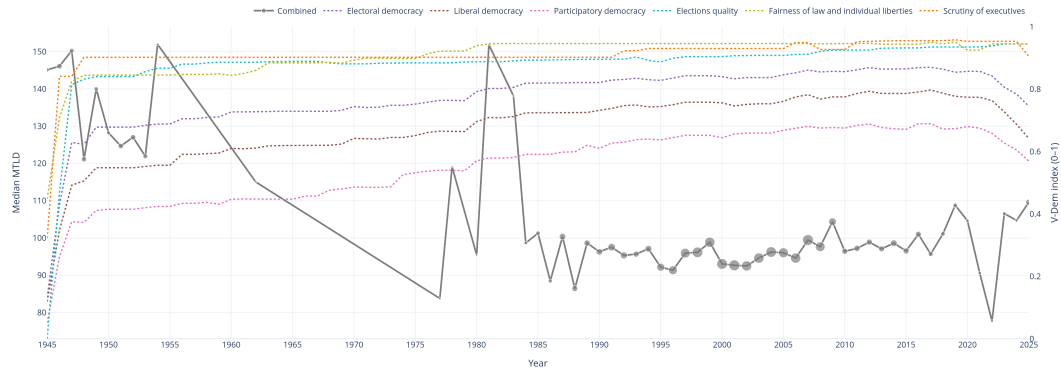
(a) MTLD values in a Boxplot by 10-year periods, with VDEM indices. Each grey box combines all speeches for all political leanings in that 10-year period.



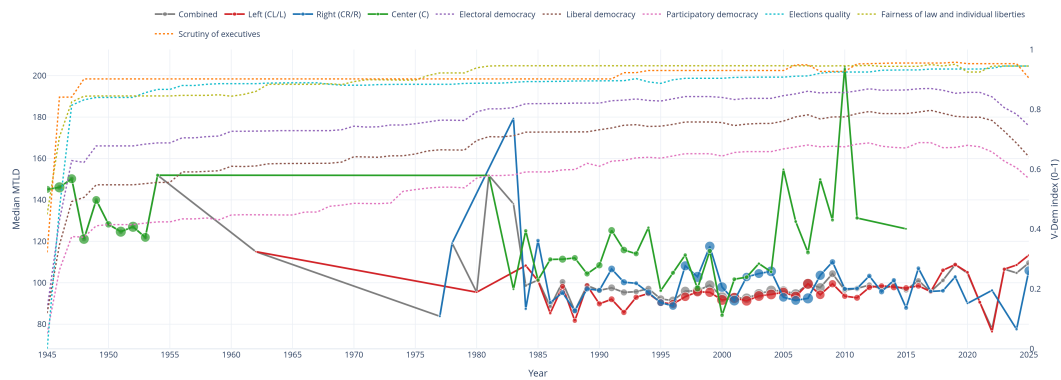
(b) MTLD values in a Boxplot by 10-year periods, with VDEM indices. As above, each grey box combines all speeches for all political leanings in that 10-year period, while the colored boxes represent respectively center-leaning (in green), left-leaning (in red), and right-leaning (in blue) MTLD values in that 10-year period.

Figure 6: MTLD values in a Boxplot by 10-year periods. There seem to be a reduction in MTLD variance starting from 1989 all the way to 2025. If this is a correlation with the inflection of some VDEM indices, it is not clear.

4.1.3.3 ALL



(a) MTLD values plotted for each year, aggregating all speeches from all political leanings. Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The MTLD line chart is plotted against VDEM indices.



(b) MTLD values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The MTLD line chart is plotted against VDEM indices.

Figure 7: MTLD values aggregated for each year, plotted against VDEM indices. There does not seem to be a correlation between a decrease in complexity and a decrease in democratic indices. Instead, the graph seems to suggest the opposite.

4.1.4 CONCLUSIONS

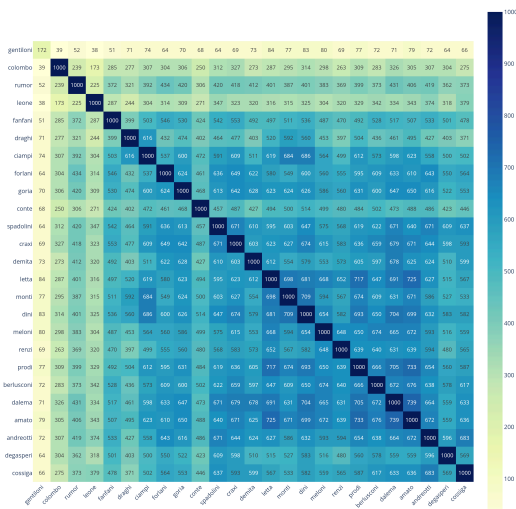
There does not seem to be a clear correlation between MTLD values and democratic indices, with the exception for the variance which ...



4.2 POLARIZATION

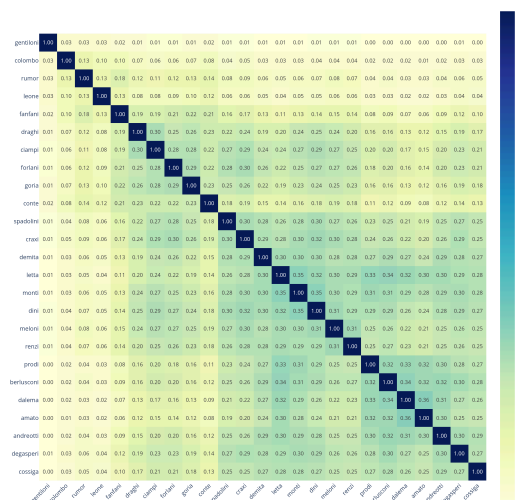
4.2.1 PRIME MINISTER'S LEXICAL SIMILARITY

Top-1000 Overlap Matrix (words in common)



(a) Words similarity.

Jaccard Similarity Matrix



(b) Jaccard similarity.

Figure 8: Heatmaps showing lexical similarity between Prime Ministers' speeches. 8a was made by pooling the 1000 most used words by each Prime Minister, then calculating how many words are shared between each of them. 8b, on the other hand, is made by calculating the Jaccard index between each pair of Prime Ministers, by pooling *all* words from all speeches used by each Prime Minister.

4.2.2 POLITICAL LEANINGS LEXICONS OVER TIME

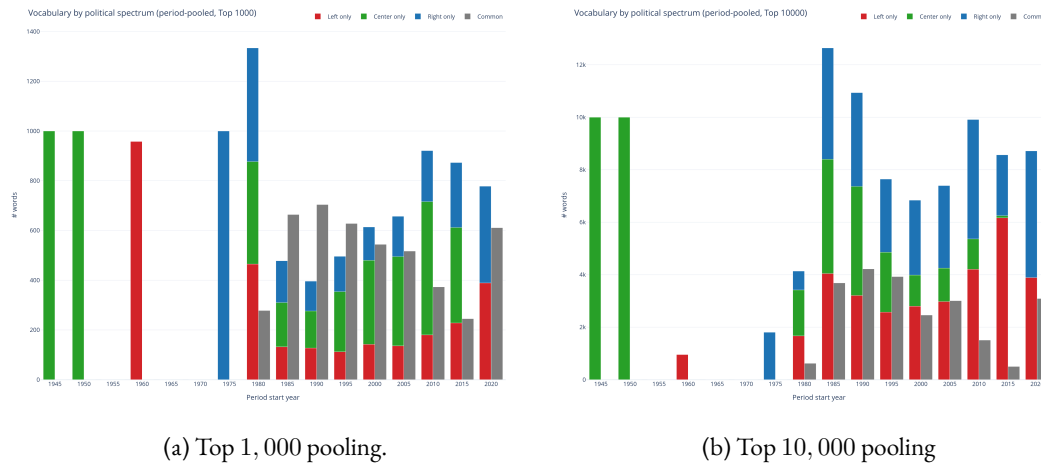


Figure 9: All speeches are pooled in 5-year time frames. Then, for each 5-year time frame, the top-N most frequently used words are extracted for each political leaning¹¹ (one thousands words for graph at 9a, and ten thousands words for graph at 9b). This results in three distinct sets¹² of words for left, right, and center leaning speeches respectively. The graph shows, for each 5-year time frame, words in common among all leanings (in grey), and words unique for each leaning (in red, blue, and green respectively). For the years with most data (from 1985 to 2015), 9a shows a progressive increase in unique vocabularies compared to a decrease in shared ones, which can be interpreted as symptoms of increasing polarization.

¹¹A speech is considered “right-leaning” if it pertains a right-leaning Prime Minister.

¹²When speeches for each leaning are present. Otherwise only two, one, or no set at all can be extracted from that 5-year time frame.

4.2.3 T-SNE

4.2.3.1 WORD2VEC: WORD EMBEDDINGS

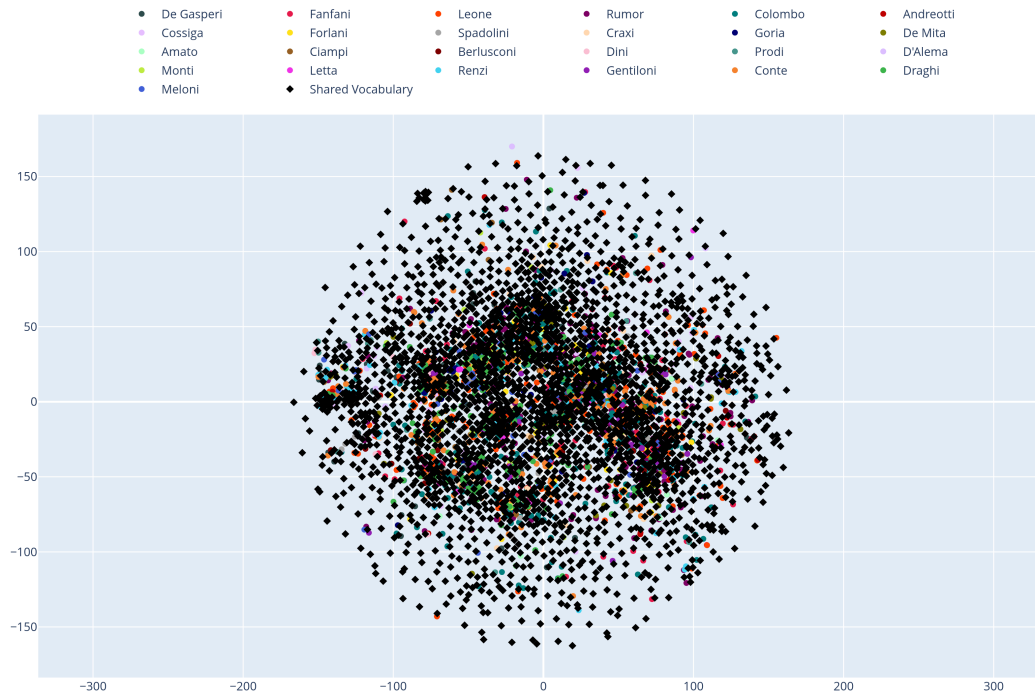


Figure 4.10: Top-1000 words¹³ from each Prime Minister. The embedding vectors are pretrained Italian word vectors (CBOW, 300 dimensions) taken from *fastText*. Words shared by all Prime Ministers are displayed as black rhombuses. The graph shows a lot of clustering, suggesting a certain similarity among Prime Ministers. That being said, the graph is also too hard to read to gain any conclusive evidence.

,

¹³Top-1000 words means pooling the one thousand most used words of a specific Prime Minister. Stopwords are removed and the lexicon is lemmatized.

4.2.3.2 Doc2Vec: SPEECH EMBEDDINGS

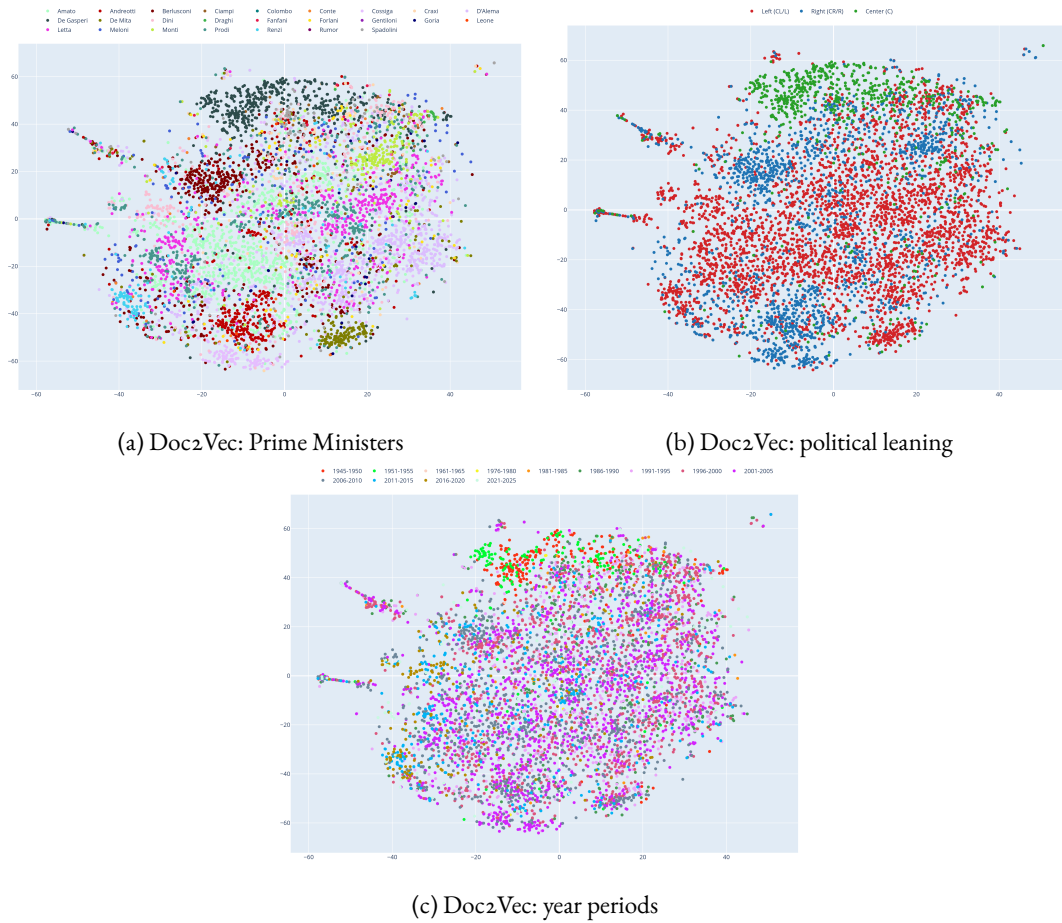


Figure 11: Using `TfidfVectorizer` from `sklearn` library, I computed one embedding for each speech¹⁴. The embeddings can then be projected in two dimensions. 11a shows the projected speech vectors grouped by Prime Ministers. We can see clustering, suggesting that most prime ministers have a lexical style and wording that are unique to them. 11b groups the vectors by political leaning, showing that different leanings occupy different and distinct regions in the vector space. Both graphs strongly hints to a polarized political landscape.

On the other hand, 11c shows two main clustering regions: on the top, speeches from 1945 to 1955 gather, while speeches from 1991 to 2025 are more uniformly scattered across the whole vector space. This suggests that earlier speeches were significantly different compared to modern ones, while speeches from the nineties onward are quite similar.

TODO: compare with VDEM polarization indices (if exists).

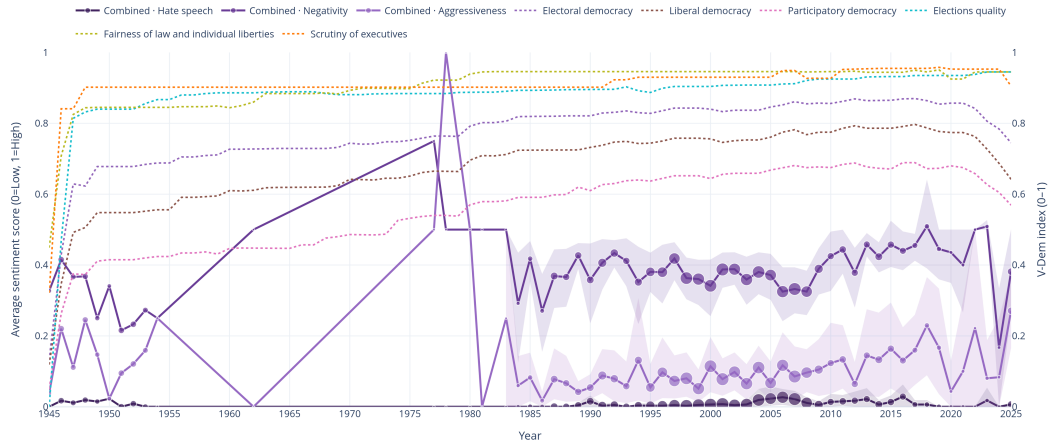
4.2.4 CONCLUSIONS

We see quite a bit of polarization, but ...

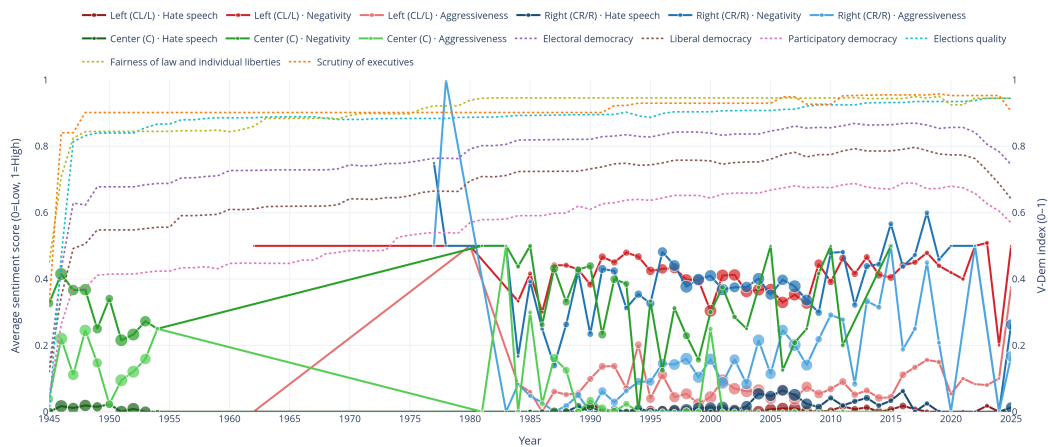
¹⁴Each speech counts as a document.

4.3 SENTIMENT ANALYSIS

4.3.1 DEMOCRATIC INDICES AGAINST SENTIMENT SCORING



(a) Sentiment values plotted for each year, aggregating all speeches from all political leanings. Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The confidence intervals are shown. The sentiment scores chart is plotted against VDEM indices.



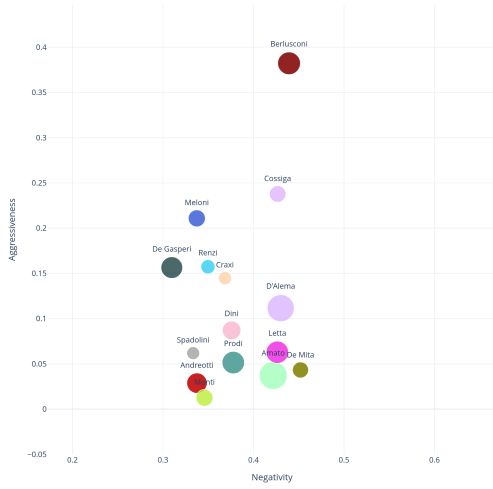
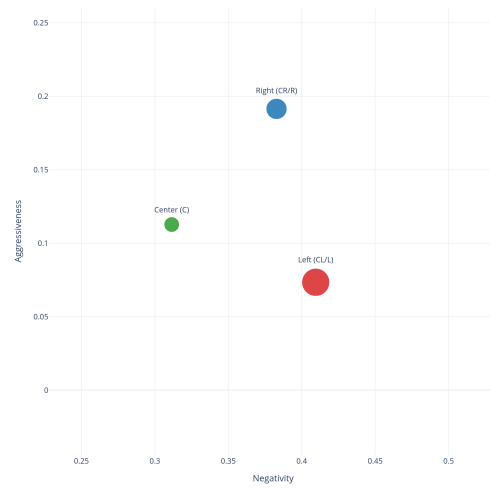
(b) Sentiment values plotted for each year, aggregating speeches from three political leanings, respectively colored as green (center), red (left), and blue (right). Each dot of the graph (for each year) is scaled based on the amount of speeches available for that specific year. The The sentiment scores line chart is plotted against VDEM indices.

Figure 12: Sentiment analysis results for *hate speech*, *aggressiveness*, and *negativity* plotted against VDEM indices.

Its hart to tell wether there is any correlation between them.

4.3.2 TOXICITY (TODO: CHANGE NAME)

First thing first: hate speech is so low for everyone that we can ignore it


(a) Sentiment: Prime Ministers ($|\text{speeches}| \geq 100$).


(b) Sentiment: leanings.



(c) Sentiment: years zoomed out.



(d) Sentiment: years zoomed in.

Figure 13: Sentiment analysis results for *negativity* and *aggressiveness*. Values go from zero to one, and the classification task considers three possible values for each speech: *low* (0), *mid* (0.5), and *high* (1).

Size of bubbles depends on the amount of speeches. 13a project results for single Prime Ministers: Berlusconi stands out as the most aggressive by far. 13b project results aggregated by leanings. The right appears to be more aggressive, while the left more negative. The center seems to be more positive than both. The last two figures show the sentiment scores aggregated by years (with 5-year pool range). While 13c shows an outlier for the period 1976 – 1980, 13d shows that overall the sentiment scores are quite close to each other, without great variance especially for periods with a lot of speeches.

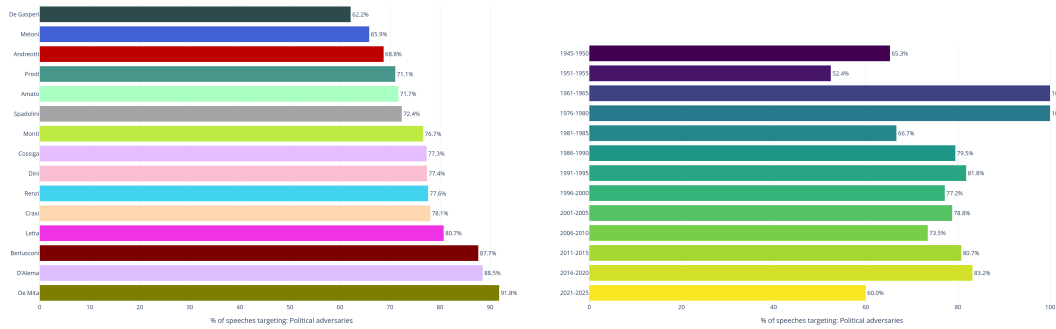
4.3.3 TARGETS

Five possible targets:

- None;
- Political adversaries;
- Gender minorities;

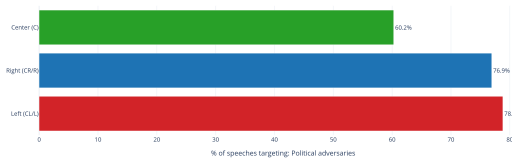
- Religious minorities;
- Ethnic minorities.

The targets are classified independently to the sentiment scores. For example, a speech could have *low* aggressiveness and still be targeting political adversaries.



(a) Target: political adversaries. Grouped by: PMs.

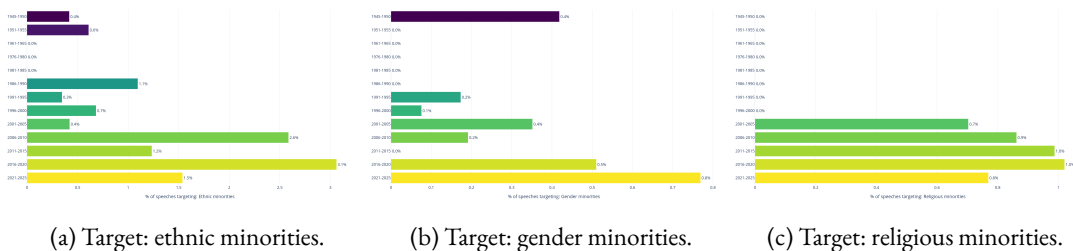
(b) Target: political adversaries. Grouped by: years.



(c) Target: political adversaries. Grouped by: leanings.

Figure 14: All graphs show the percentage of speeches that target political adversaries. 14a shows which Prime Minister speaks more often about the opposition (prime ministers with less than 100 speeches are omitted). 14b shows in which 5-year periods there are more speeches targeting political adversaries (ordered by year). Lastly, 14c shows which of the three political fields is more likely to speak about the opposing front. The results show how, across the board, more than one speech out of two targets the opposition. This is true regardless of time period or affiliation.

Thankfully, Italian Prime Minister apparently do not target minorities often.



(a) Target: ethnic minorities.

(b) Target: gender minorities.

(c) Target: religious minorities.

Figure 15: Overall, an extremely low percentage of speeches target minority groups. Therefore, I will ignore this analysis. (TODO explain better.. within error yada yada).

4.3.4 CONCLUSIONS

Italian prime minister seem to be overall a bit aggressive and negative, and they speak more often then not about the opposition. Almost no traces of hate speech and attacks against minority groups are found in the data.

This does not seem to change over time, failing to explain the VDEM indices drop in 2021.







5

Discussion & Future Works

- Analyses results
 - so can we explain VDEM drops and stuff?
- How to improve the work
 - more scraping (increase dataset)
 - better transcription (even tho they are pretty good)
 - bigger LLMs for the sentiment analysis
 - better doc2vec? gensim still not compatible with current Python version
- How to expand the work
 - include leader of opposition
 - include trend setter of the time
 - include other countries (eg France or USA)

A

Appendix

I.1 TABLES

I.1.1 NOT COLORED

Level	Field	Description	Example
1	tenant_id	Logical isolation between organizations	agroTech01
2	site_type	Type of agricultural facility	greenhouse, open field
3	site_id	Physical site identifier	GH-001
4	section_id	Section within the site	section-2
5	zone_id	Operational zone	zone-A
6	entity_type	Device category	sensor, actuator
7	entity_subclass	Specific device type	temp, ph, water pump
8	entity_id	Unique device identifier	mqtt-device-03
9	message_type	Message classification	data, cmd, status, alert
10	target_id	Target device for commands (optional)	cooler-001

Table A.2: 10-level topic hierarchy structure

I.I.2 COLORED

Level	Field	Description	Example
1	tenant_id	Logical isolation between organizations	agroTech01
2	site_type	Type of agricultural facility	greenhouse, open field
3	site_id	Physical site identifier	GH-001
4	section_id	Section within the site	section-2
5	zone_id	Operational zone	zone-A
6	entity_type	Device category	sensor, actuator
7	entity_subclass	Specific device type	temp, ph, water pump
8	entity_id	Unique device identifier	mqtt-device-03
9	message_type	Message classification	data, cmd, status, alert
10	target_id	Target device for commands (optional)	cooler-001

Table A.3: 10-level topic hierarchy structure

I.I.3 HALF COLORED HALF NOT

Level	Field	Description	Example
1	tenant_id	Logical isolation between organizations	agroTech01
2	site_type	Type of agricultural facility	greenhouse, open field
3	site_id	Physical site identifier	GH-001
4	section_id	Section within the site	section-2
5	zone_id	Operational zone	zone-A
6	entity_type	Device category	sensor, actuator
7	entity_subclass	Specific device type	temp, ph, water pump
8	entity_id	Unique device identifier	mqtt-device-03
9	message_type	Message classification	data, cmd, status, alert
10	target_id	Target device for commands (optional)	cooler-001

Table A.4: 10-level topic hierarchy structure

I.1.4 STACKED

Table *iot_records_tenant_01*

Time	Device ID	Tenant ID	Type	Data (JSONB)
2025-11-10 08:03:12 UTC	temp_001	tenant_01	data	{"value": 22.4, "unit": "C"}
2025-11-10 08:05:44 UTC	fan_003	tenant_01	status	{"running": true, "speed": "medium"}
2025-11-10 08:07:01 UTC	temp_001	tenant_01	alert	{"code": "OVER_THRESHOLD", "value": 29.1}
2025-11-10 08:09:33 UTC	fan_003	tenant_01	cmd	{"action": "set_speed", "target": "high"}

Table *iot_records_tenant_02*

Time	Device ID	Tenant ID	Type	Data (JSONB)
2025-11-10 08:10:22 UTC	temp_007	tenant_02	data	{"value": 18.9, "unit": "C"}
2025-11-10 08:10:25 UTC	air_008	tenant_02	data	{"value": 412, "unit": "ppm"}
2025-11-10 08:12:01 UTC	temp_007	tenant_02	status	{"online": true, "battery": 87}

Table A.5: Sample rows from *iot_records_tenant_01* (top) and *iot_records_tenant_02* (bottom).

I.2 FIGURES

`#align()` without any arguments defaults to *center*.

I.2.1 SINGLE FIGURE

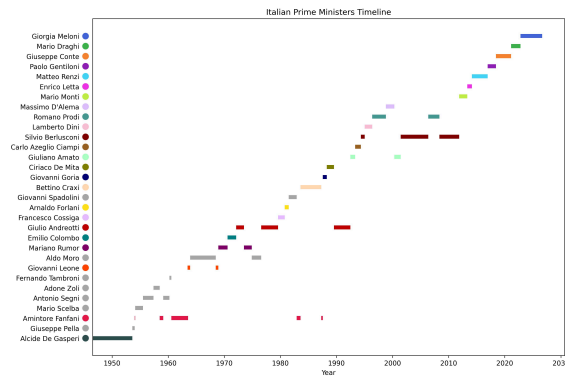


Figure A.16: Architecture of IoT (A: 3-layers) (B: 5-layers)

I.2.2 DOUBLE FIGURES

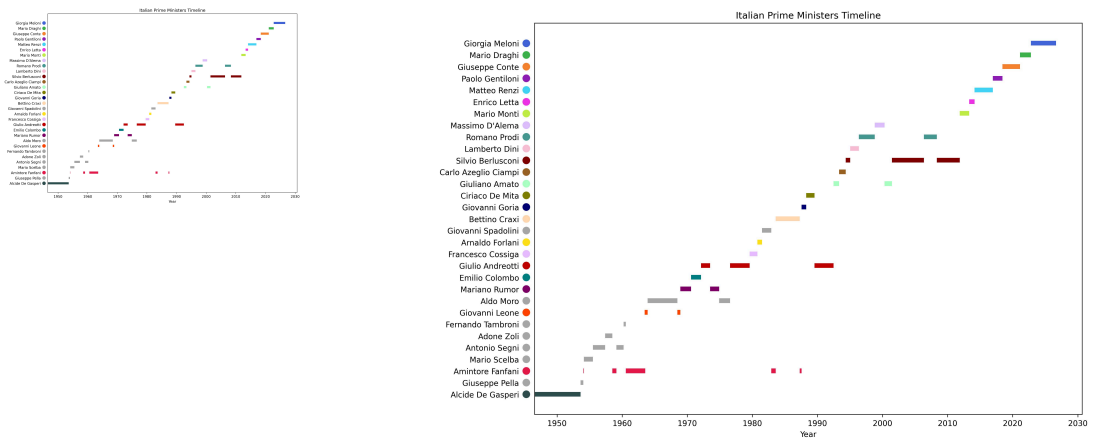


Figure A.17: Comparison between (a) Request-Response and (b) Publish-Subscribe paradigms

I.3 CODE LISTINGS

```
1 {
2   "entity": { "type": "tenant", "id": "agroTech01" },
3   "relation": "admin",
4   "subject": { "type": "user", "id": "john_doe" }
5 }
```

Code A.1: Example of a Permify tuple

```
1 def extract_page_date(soup: BeautifulSoup) -> tuple[date | None, str]:
2     """Page-level event date (meta > title > stamp > sommario prose)."""
3     for attr, val in [{"name", "dcterms.date"}, ("property", "article:published_time")]:
4         tag = soup.find("meta", {attr: val})
5         if tag and tag.get("content"):
6             d = _try_iso(tag["content"])
7             if d: return d, "meta_tag"
8
9     title = soup.find("title")
10    if title:
11        d = _try_dotted(title.get_text())
12        if d: return d, "page_title"
13
14    page_text = soup.get_text(" ", strip=True)
15    d = _try_short(page_text)
16    if d: return d, "page_stamp"
17
18    d = _try_long(page_text)
19    if d: return d, "page_sommario"
20
21    return None, "not_found"
```

Code A.2: Example of a Permify tuple

```

1  entity platform {
2      relation super_admin @user
3      permission manage_tenants = super_admin
4  }
5
6  entity tenant {
7      relation admin      @user | @user_group#member
8      relation maintainer @user | @user_group#member
9      relation member     @user | @user_group#member
10
11     permission is_admin      = admin
12     permission is_maintainer = maintainer
13     permission is_member     = member
14
15     permission manage_structure = is_admin or is_maintainer
16     permission view_structure  = manage_structure or is_member
17 }
18
19 entity macro_section {
20     [...]
21 }
22
23 entity section {
24     [...]
25 }
26
27 entity device_type {
28     attribute can_pub_data  boolean
29     attribute can_issue_cmd boolean
30     // ... other capability attributes
31
32     permission allowed_pub_data  = can_pub_data
33     permission allowed_issue_cmd = can_issue_cmd
34     // ... other permissions derived from attributes
35 }
36
37 entity device {
38     relation section      @section
39     relation type         @device_type
40     relation controller @user | @user_group#member
41
42     permission publish_data = type.can_pub_data
43     permission subscribe_cmd = type.can_sub_cmd
44     permission receive_cmd  = section.operate_devices or controller
45     permission read_data    = section.view_data
46 }

```

Code A.3: Permify schema example

```

1  reqBody := PermifyCheckRequest{
2      Metadata: PermifyMetadata{
3          SnapToken: "",
4          SchemaVersion: "",
5          Depth: 100,
6      },
7      Entity: PermifyEntity{
8          Type: entityType,
9          ID: entityID,
10     },
11
12     Permission: permission,
13
14     Subject: PermifySubject{
15         Type: subjectType,
16         ID: subjectID,
17     },
18 }

```

Code A.4: Example of a permission check request to Permify

```

1  vernemq2:
2    image: vernemq-from-source:latest
3    container_name: vernemq2
4    labels:
5      - "traefik.enable=true"
6
7      # MQTT TCP (port 1883) - Load balanced
8      - "traefik.tcp.routers.mqtt.entrypoints=mqtt"
9      - "traefik.tcp.routers.mqtt.rule=HostSNI('*')"
10     - "traefik.tcp.routers.mqtt.service=mqtt-backend"
11
12     # MQTT TLS (:8883)
13     - "traefik.tcp.routers.mqtt-secure.entrypoints=mqtts"
14     - "traefik.tcp.routers.mqtt-secure.rule=HostSNI('*')"
15     - "traefik.tcp.routers.mqtt-secure.tls=true"
16     - "traefik.tcp.routers.mqtt-secure.tls.certresolver=letsencrypt"
17     - "traefik.tcp.routers.mqtt-secure.service=mqtt-backend"
18
19     environment:
20       - DOCKER_VERNEMQ_ALLOW_ANONYMOUS=off
21       # Disable built-in ACLs to use only the webhook authentication/authorization
22       - DOCKER_VERNEMQ_PLUGINS__VMQ_ACL=off
23       # plugin webhooks managed via vernemq.confs
24       - DOCKER_VERNEMQ_PLUGINS__VMQ_WEBHOOKS=on
25       # Webhooks pointing to auth-service
26       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_AUTH__HOOK=auth_on_register
27       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_AUTH__ENDPOINT=http://auth-service:8080/auth_on_
28     register
29       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_PUBLISH__HOOK=auth_on_publish
30       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_PUBLISH__ENDPOINT=http://auth-service:8080/auth_on_
31     publish
32       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_SUBSCRIBE__HOOK=auth_on_subscribe
33       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__JWT_SUBSCRIBE__ENDPOINT=http://auth-service:8080/auth_on_
34     subscribe
35       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__ON_CLIENT_OFFLINE__HOOK=on_client_offline
36       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__ON_CLIENT_OFFLINE__ENDPOINT=http://auth-service:8080/on_
37     client_offline
38       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__ON_CLIENT_GONE__HOOK=on_client_gone
39       - DOCKER_VERNEMQ_VMQ_WEBHOOKS__ON_CLIENT_GONE__ENDPOINT=http://auth-service:8080/on_
40     client_gone
41
42     # auto-join cluster to first node
43     - DOCKER_VERNEMQ_DISCOVERY_NODE=${VERNEMQ_IP_1}
44
45     networks:
46       vernemq-net:
47         ipv4_address: ${VERNEMQ_IP_2}
48
49     depends_on:
50       - vernemq1

```

Code A.5: VerneMQ configuration in Docker Compose - relevant parts

```

1  func main() {
2    [...]
3
4    r.POST("/auth_on_register", authOnRegister)
5    r.POST("/auth_on_publish", authOnPublish)
6    r.POST("/auth_on_subscribe", authOnSubscribe)
7    r.POST("/on_client_offline", onClientOffline)
8    r.POST("/on_client_gone", onClientGone)
9
10   [...]
11 }

```

Code A.6: Auth service - webhook endpoints handlers

```

1  {
2  "name": "IOT_INCOMING_DATA",
3  "subjects": ["incoming.mqtt.>", "incoming.coap.>", "outcoming.coap.>", "outcoming.mqtt.>"],
4  "retention": "limits",
5  "max_consumers": -1,
6  "max_msgs": 100000,
7  "max_bytes": -1,
8  "max_age": 864000000000000,
9  "storage": "file",
10 "num_replicas": 3
11 }

```

Code A.7: NATS JetStream stream configuration

```

1  dtlsConfig := &piondtls.Config{
2  PSK: func(hint []byte) ([]byte, error) {
3      deviceHint := string(hint)
4      log.Printf("DTLS Handshake: Authenticating device '%s'", deviceHint)
5
6      key, err := pskStore.GetKey(hint)
7      if err != nil {
8          log.Printf("Authentication failed for device '%s': %v", deviceHint, err)
9          return nil, err
10     }
11
12     log.Printf("✓ Device '%s' authenticated", deviceHint)
13     return key, nil
14 },
15 PSKIdentityHint: []byte("coap-bridge-server"),
16 CipherSuites: []piondtls.CipherSuiteID{
17     piondtls.TLS_PSK_WITH_AES_128_CCM_8,
18     piondtls.TLS_PSK_WITH_AES_128_GCM_SHA256,
19 },
20 }

```

Code A.8: DTLS configuration with PSK callback function

```

1  type InputPayload struct {
2      DeviceID string      `json:"device"`
3      Type     string      `json:"type"`
4      Ts       int64       `json:"ts"`
5      Data     json.RawMessage `json:"data"`
6      TenantID string      `json:"tenant_id"`
7
8      [...]
9
10     tableName := tenantTableName(payload.TenantID)
11     // Create the table if it doesn't exist (only for 1st message)
12     if res := db.Table(tableName).Create(&dbRecord); res.Error != nil {
13         log.Printf("DB Insert Error (table %q): %v", tableName, res.Error)
14         msg.Nak()
15     } else {
16         log.Printf("Saved to %q: device=%s tenant=%s", tableName, payload.DeviceID,
17             payload.TenantID)
18         msg.Ack()
19     }
20 }

```

Code A.9: DB Writer input payload structure

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